5. Loop

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Review

for Statement

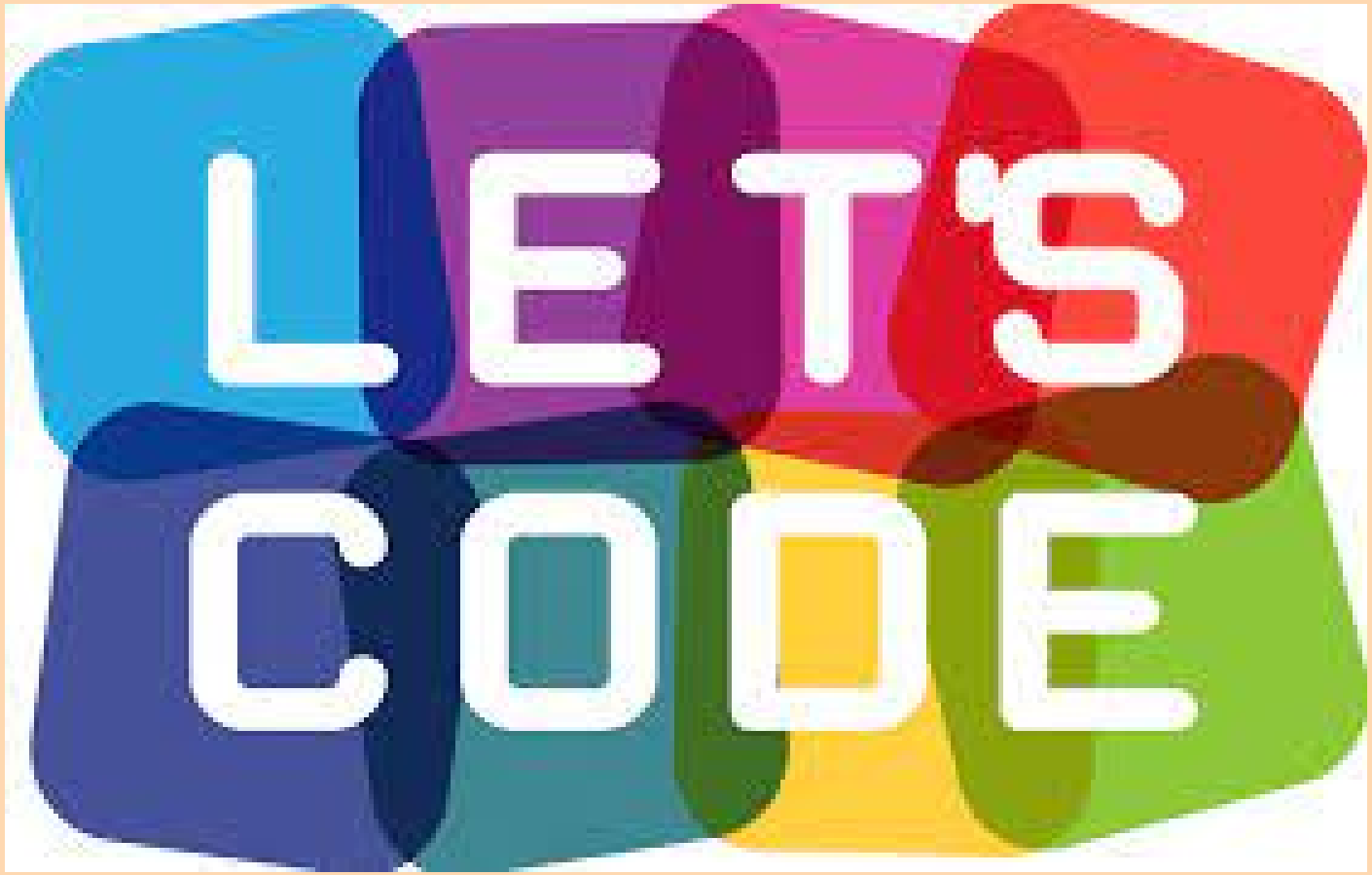
Nested Loops

Overview

Loops (More Examples)

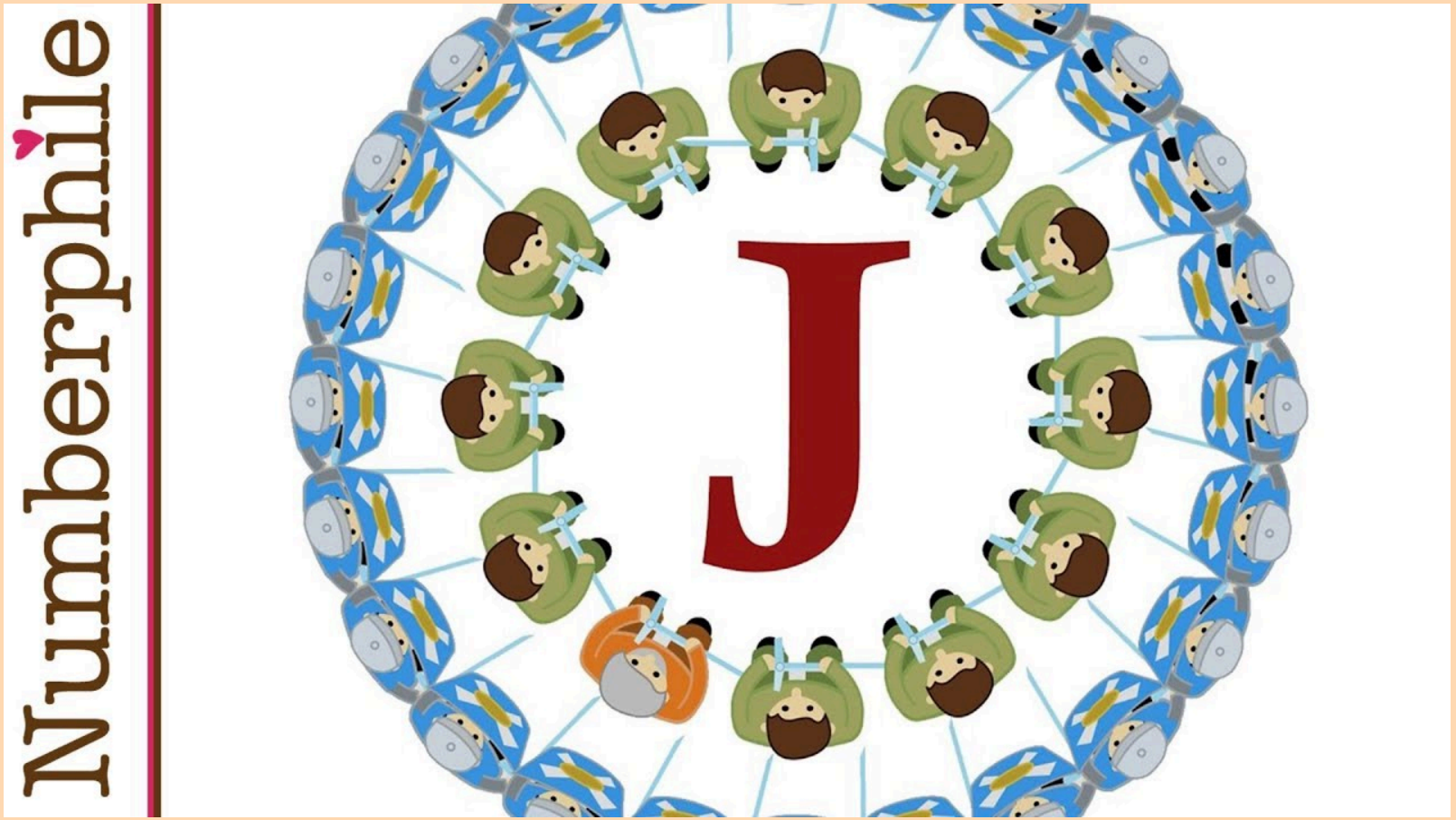
vector

Multi-dimensional Arrays (多维数组)

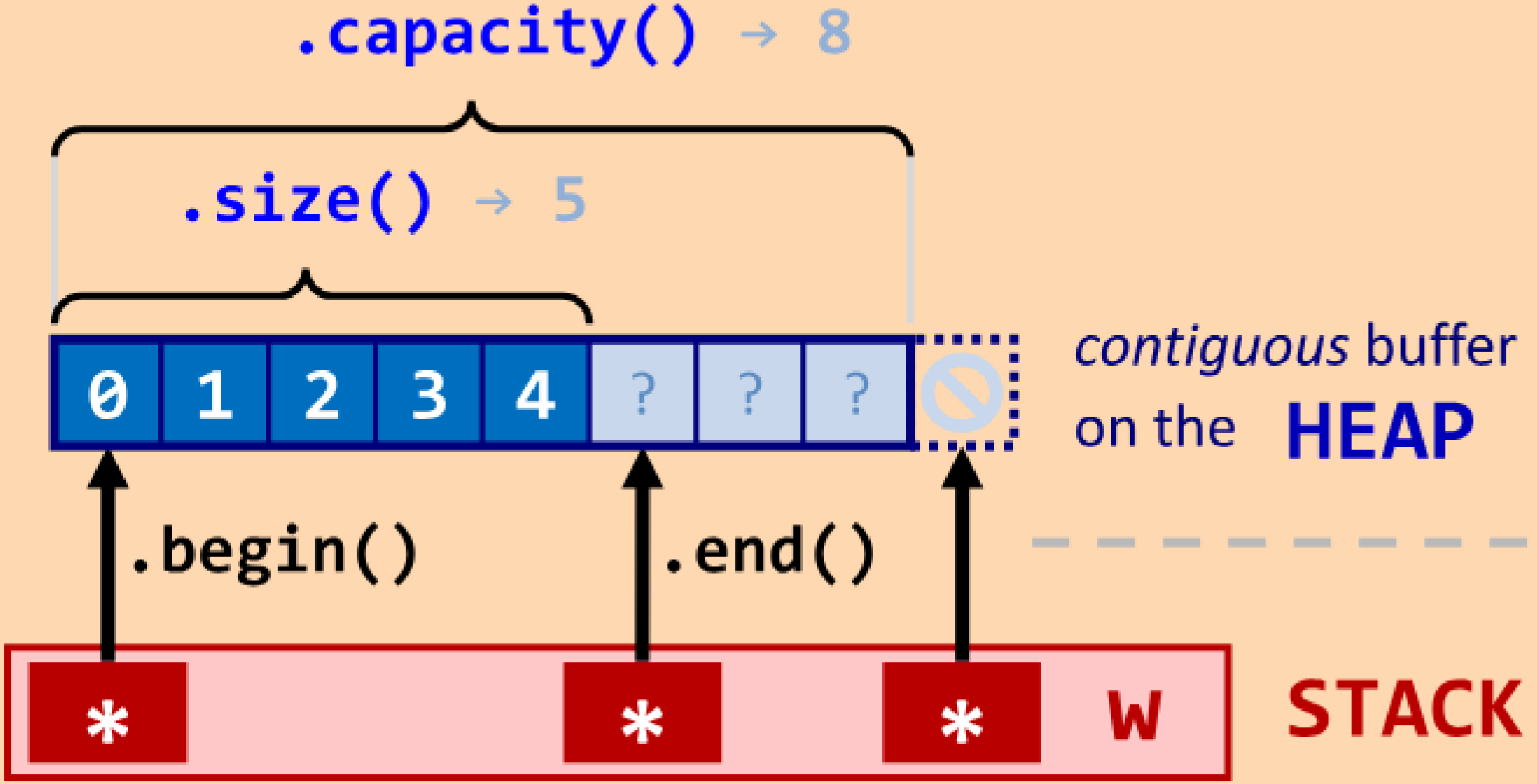


josephus.cpp game-of-life.cpp

insertion-sort.cpp bubble-sort.cpp



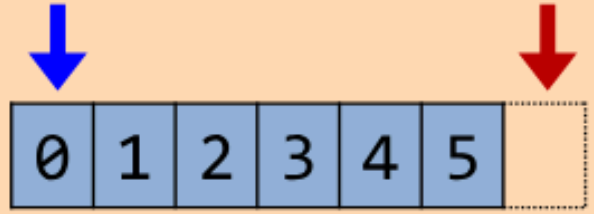
"I hate the Josephus Game!"


.capacity() → 8
.size() → 5
0
1
2
3
4
?
?
?
contiguous buffer
on the HEAP
.begin()
.end()
*
*
*
w
STACK

```
std::vector<int> v {0, 1, 2, 3, 4, 5};
```

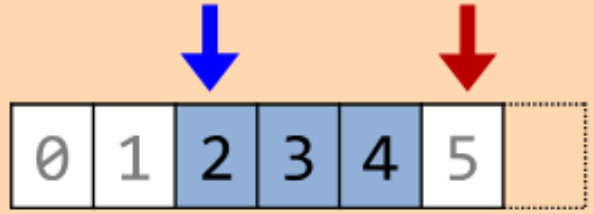
`begin(v)`, `end(v)`

6 elements



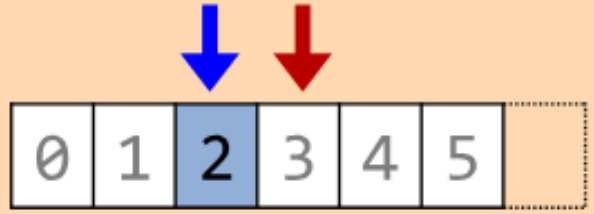
`begin(v)+2`, `begin(v)+5`

3 elements



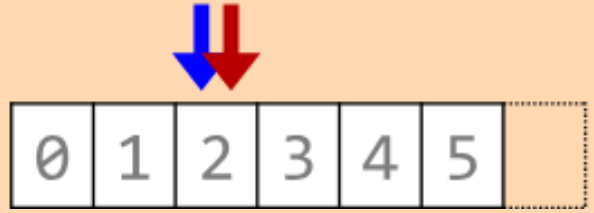
`begin(v)+2`, `begin(v)+3`

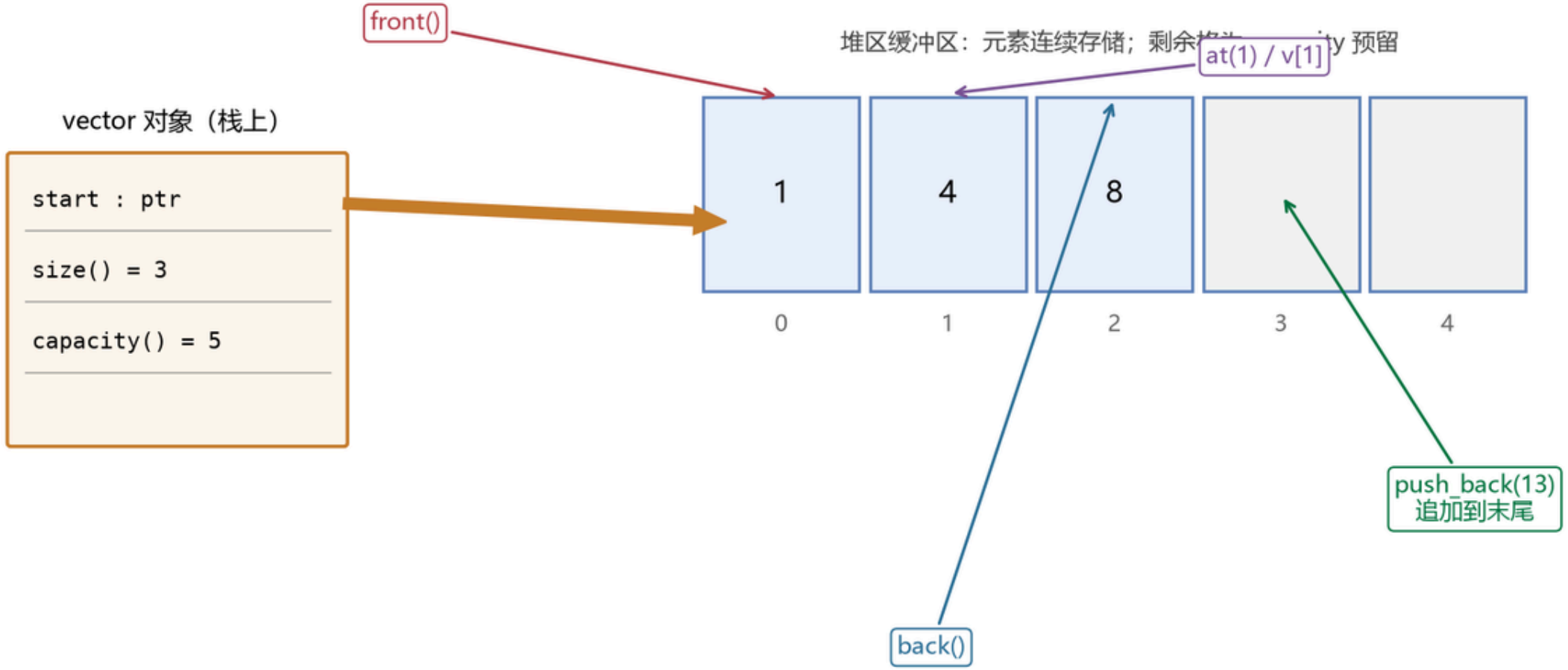
1 element

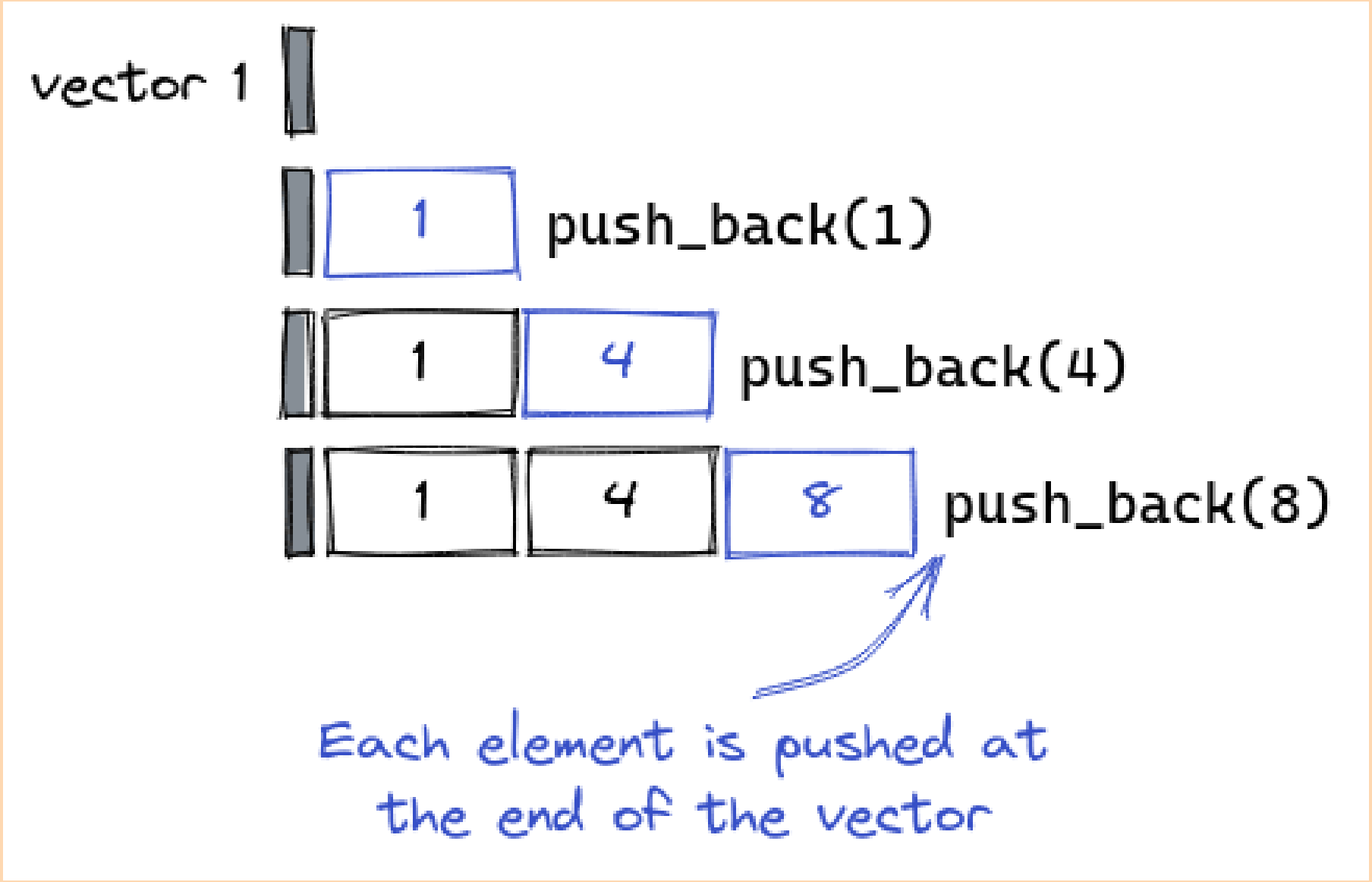


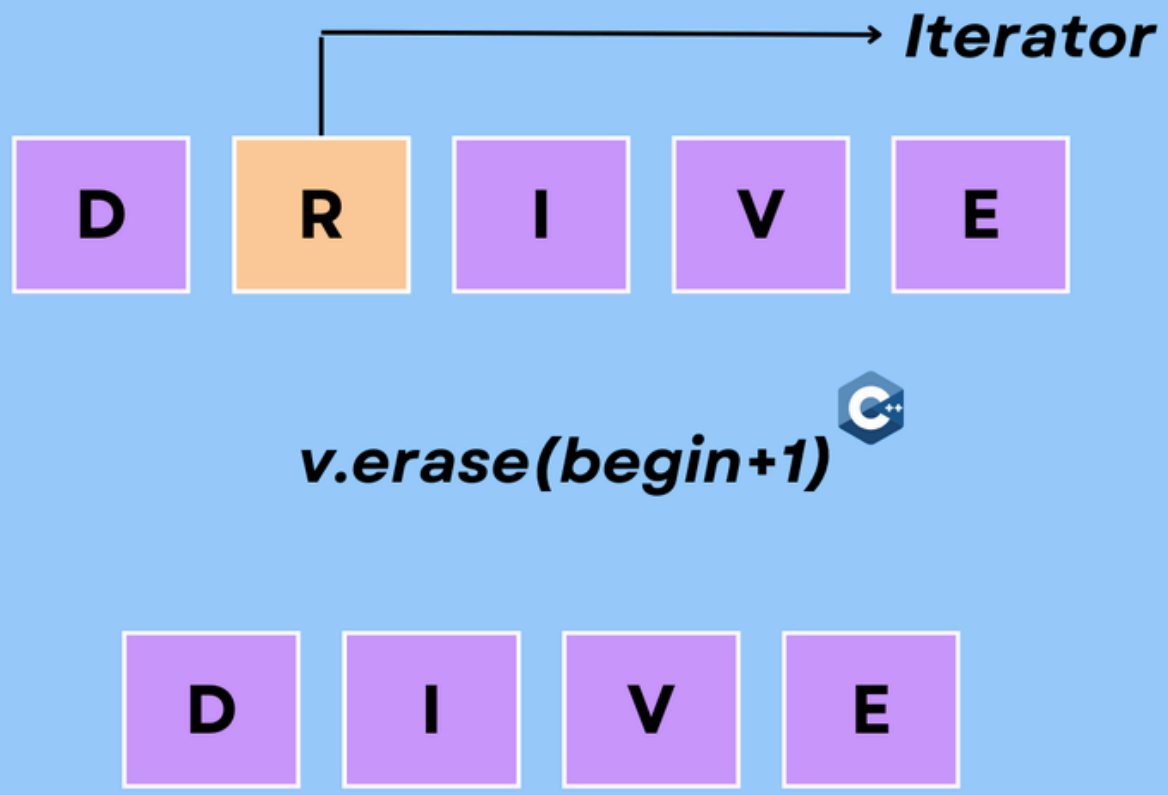
`begin(v)+2`, `begin(v)+2`

empty range

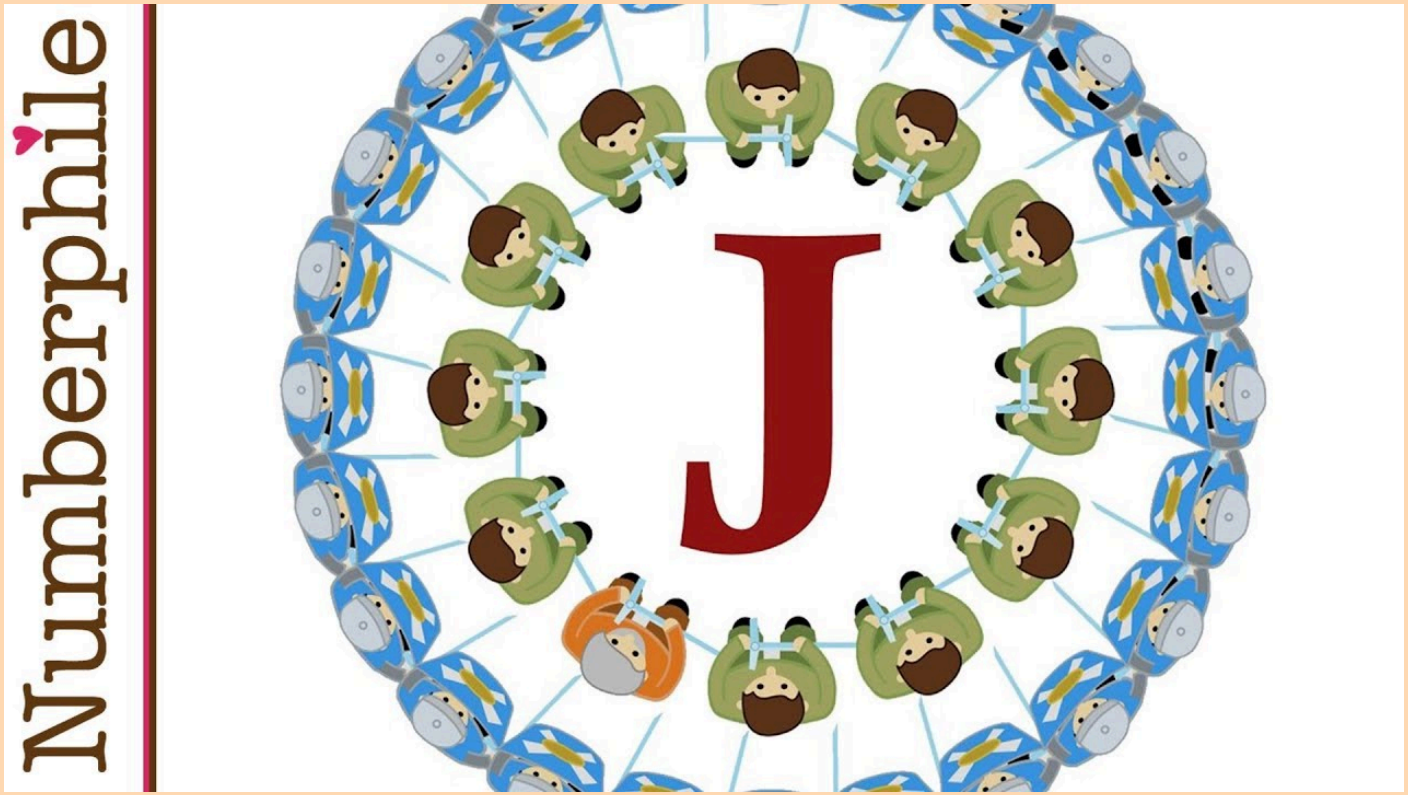



front()
堆区缓冲区：元素连续存储；剩余
y 预留
at(1) / v[1]
vector 对象（栈上）
start : ptr
size() = 3
capacity() = 5
1
4
8
0
1
2
3
4
push_back(13)
追加到末尾
back()


vector 1
1
push_back(1)
1
4
push_back(4)
1
4
8
push_back(8)
Each element is pushed at
the end of the vector


Iterator
D
R
I
V
E
v.erase(begin+1)
D
I
V
E

$$J(2^m + l) = 2l + 1 \quad (m \geq 0 \wedge 0 \leq l < 2^m)$$



$$J(12) = J(2^3 + 4) = 2 \times 4 + 1 = 9$$

Conway's Game of Life @ wiki



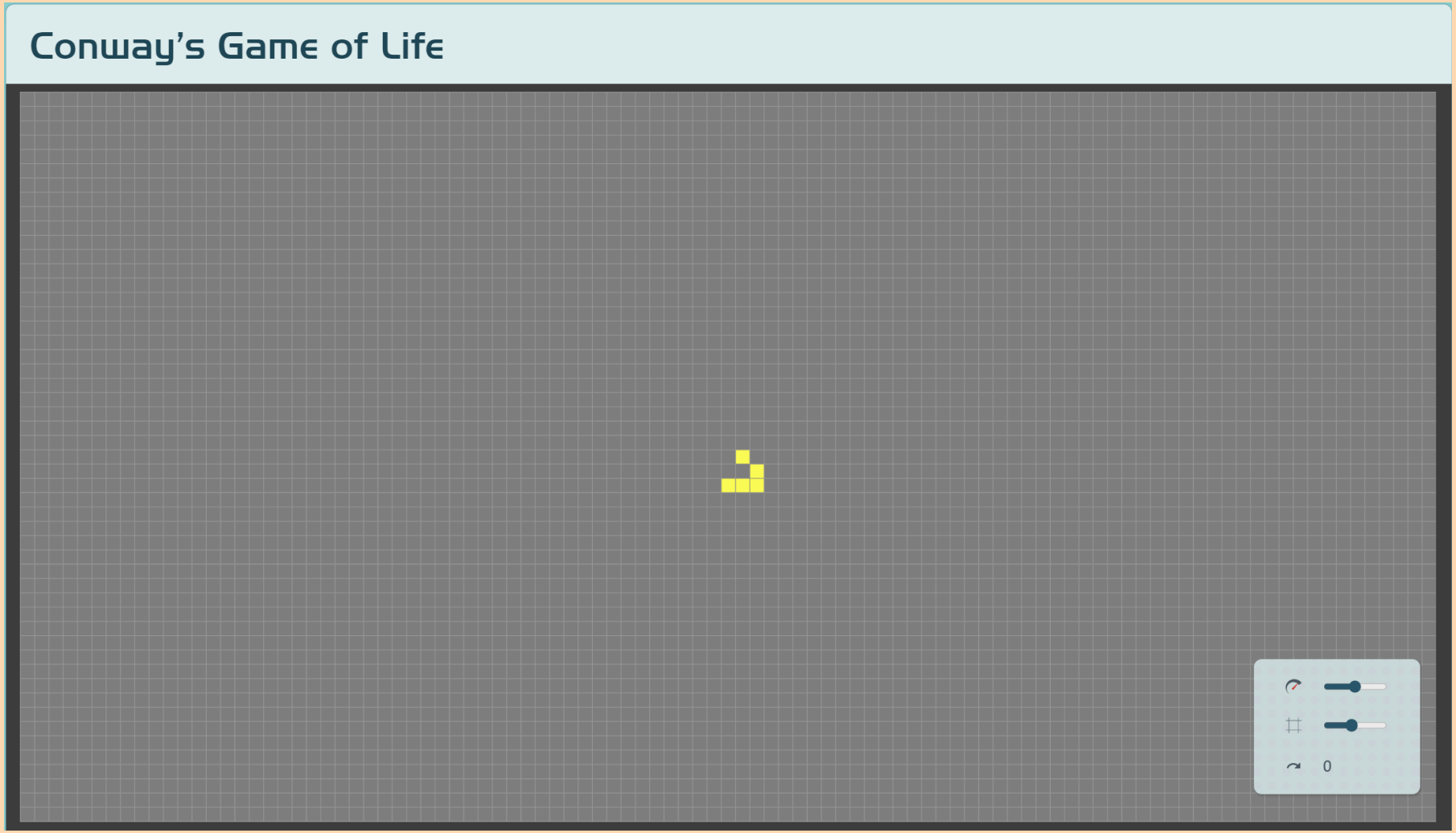
John Horton Conway ($1937 \sim 2020$)

playgameoflife.com (Cellular Automata; 元胞自动机)

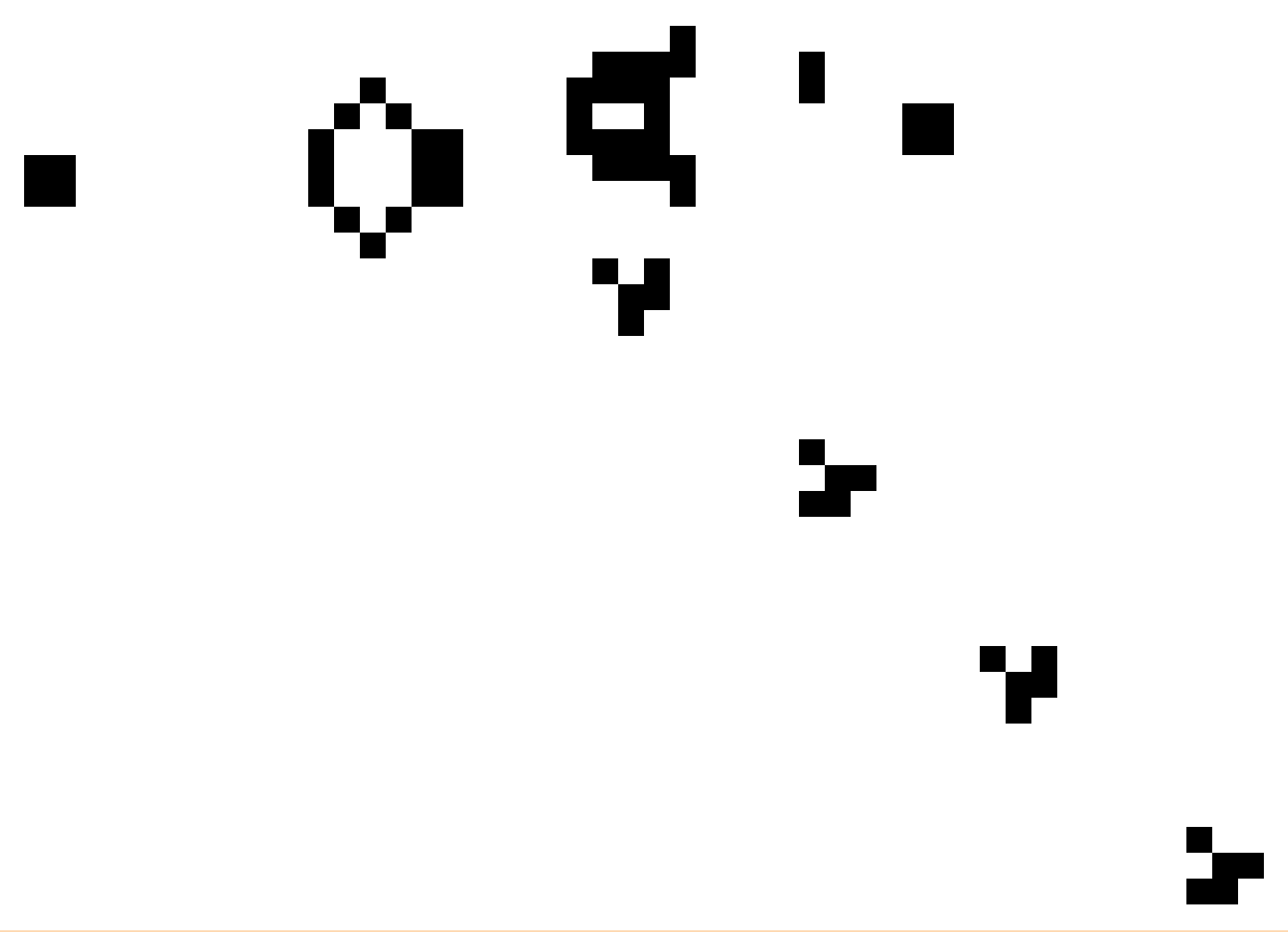
- Any **live** cell with two or three live neighbours survives.
- All other **live** cells die in the next generation.

- Any **dead** cell with three live neighbours becomes a live cell.
- All other **dead** cells stay dead.

Glider



Gosper glider gun

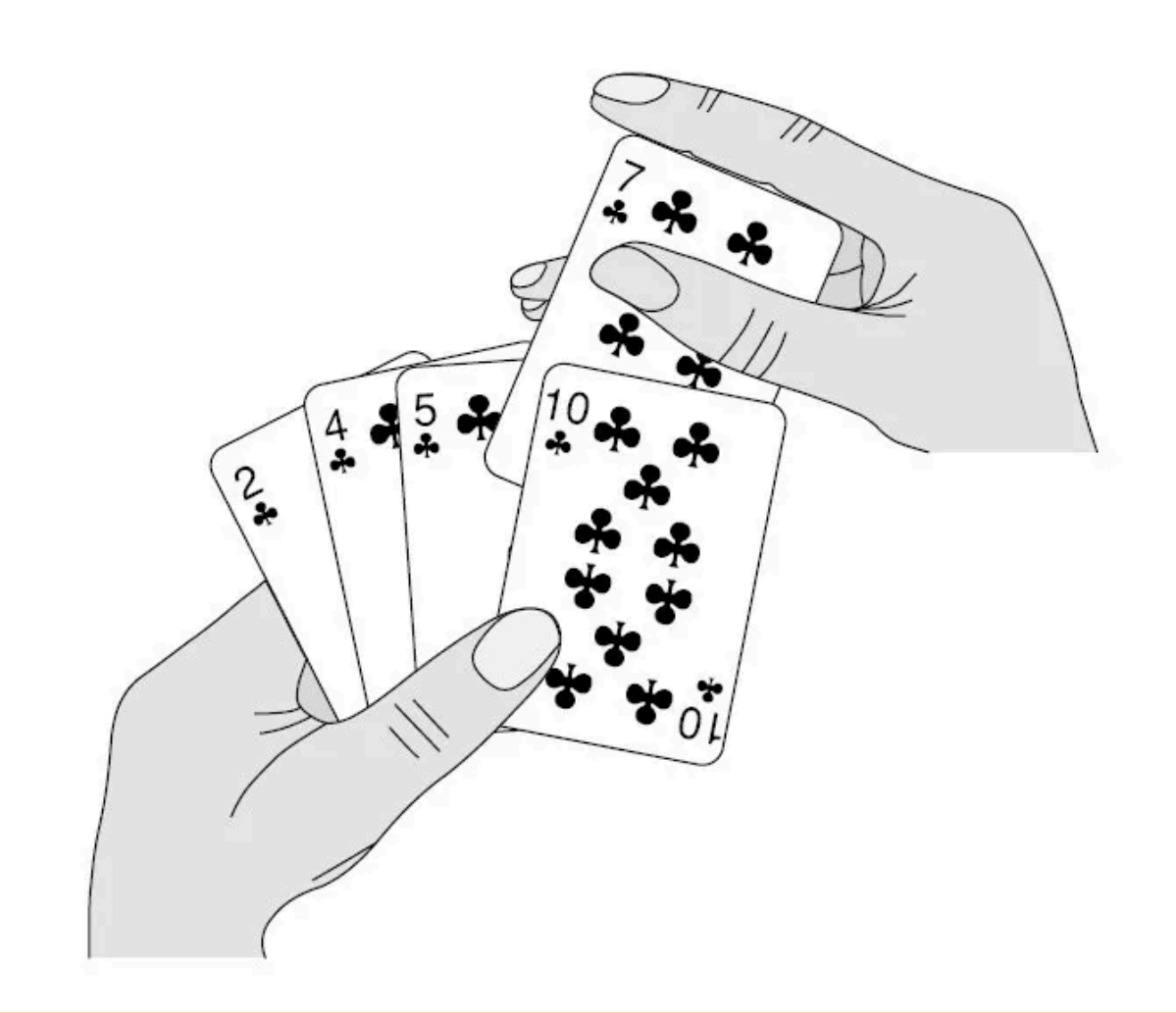


Puffer-type breeder





Insertion Sort



今日天气百一十度。打牌。

七月四日（星二）

读 Plato's "Apology of Socrates"。

今日为美国独立纪念日，夜八时至湖上观此间庆祝会。士女来游者无算，公园中百戏俱陈，小儿女燃花爆为乐。既而焰火作矣，五光十色，备极精巧。九时半始归。

七月五日（星三）

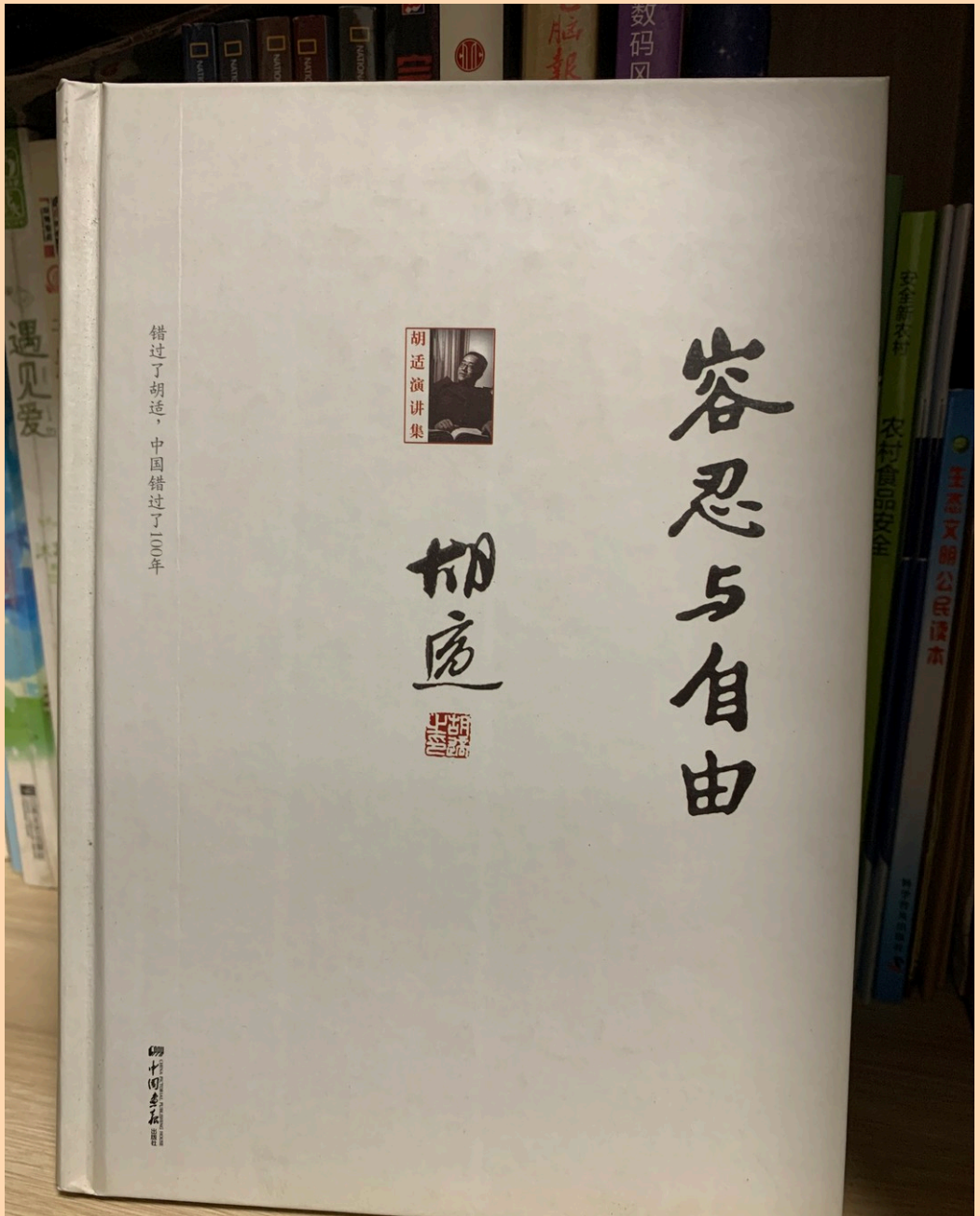
往暑期学校注册。下午打牌。

七月六日（星四）

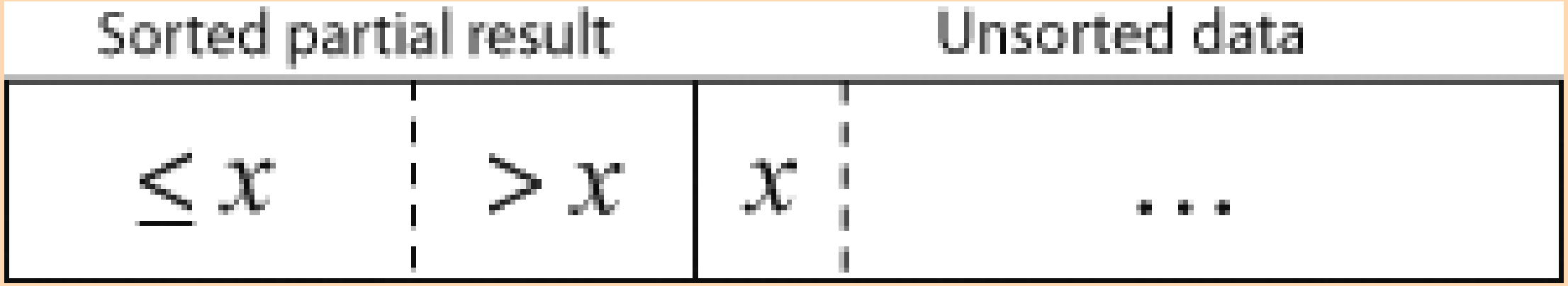
暑期学校第一日，化学（八时至一时）。打牌。

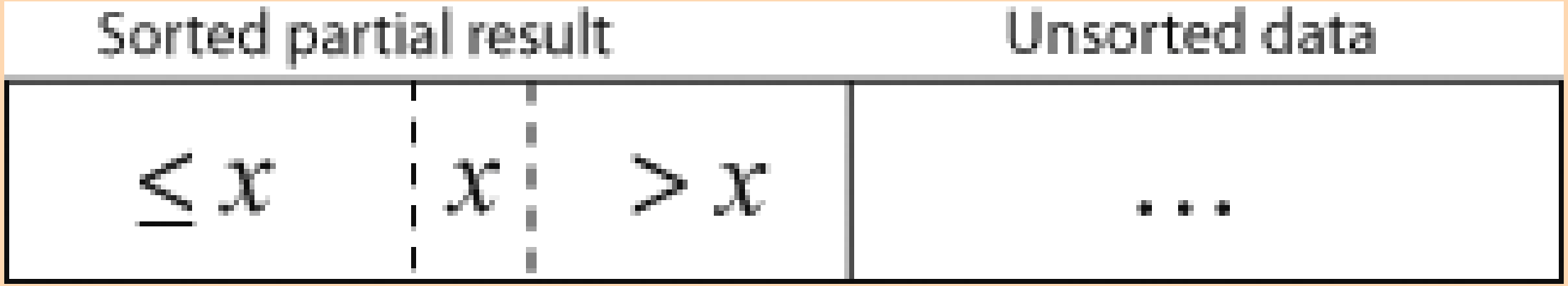
七月七日（星五）

上课。打牌。

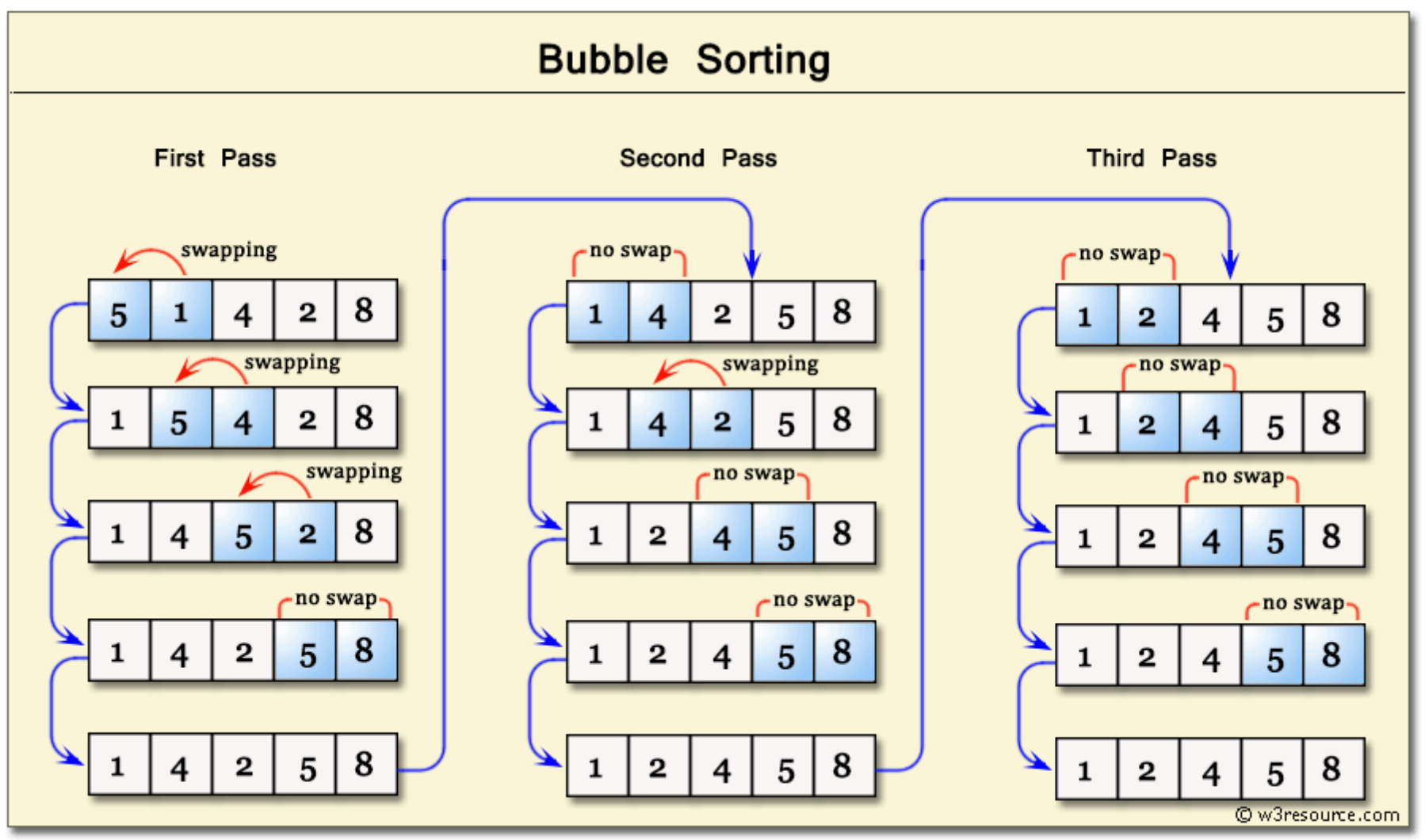


6 5 3 1 8 7 2 4


Sorted partial result
Unsorted data
$\leq x$
$> x$
x
...


Sorted partial result
Unsorted data
$\leq x$
x
$> x$
...

Bubble Sort


Bubble Sorting
First Pass
Second Pass
Third Pass
swapping
no swap
5 1 4 2 8
1 5 4 2 8
1 4 5 2 8
1 4 2 5 8
1 4 2 5 8
1 4 2 5 8
1 2 4 5 8
1 2 4 5 8
1 2 4 5 8
1 2 4 5 8
1 2 4 5 8
1 2 4 5 8
1 2 4 5 8
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6 5 3 1 8 7 2 4


THANK YOU!